Project 4: AWS Route 53 Domain Routing

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1. Project Overview & Objectives

The core objective of this project is to implement DNS domain routing using **AWS Route 53**. The project demonstrates how to map a professionally registered custom domain name to host websites using two distinct infrastructural methods:

- **Method 1:** Pointing the domain to a serverless static website hosted inside an **AWS S3 Bucket**.
- **Method 2:** Pointing the domain directly to a live web server configured on an **AWS EC2 Instance** using a static **Elastic IP**.

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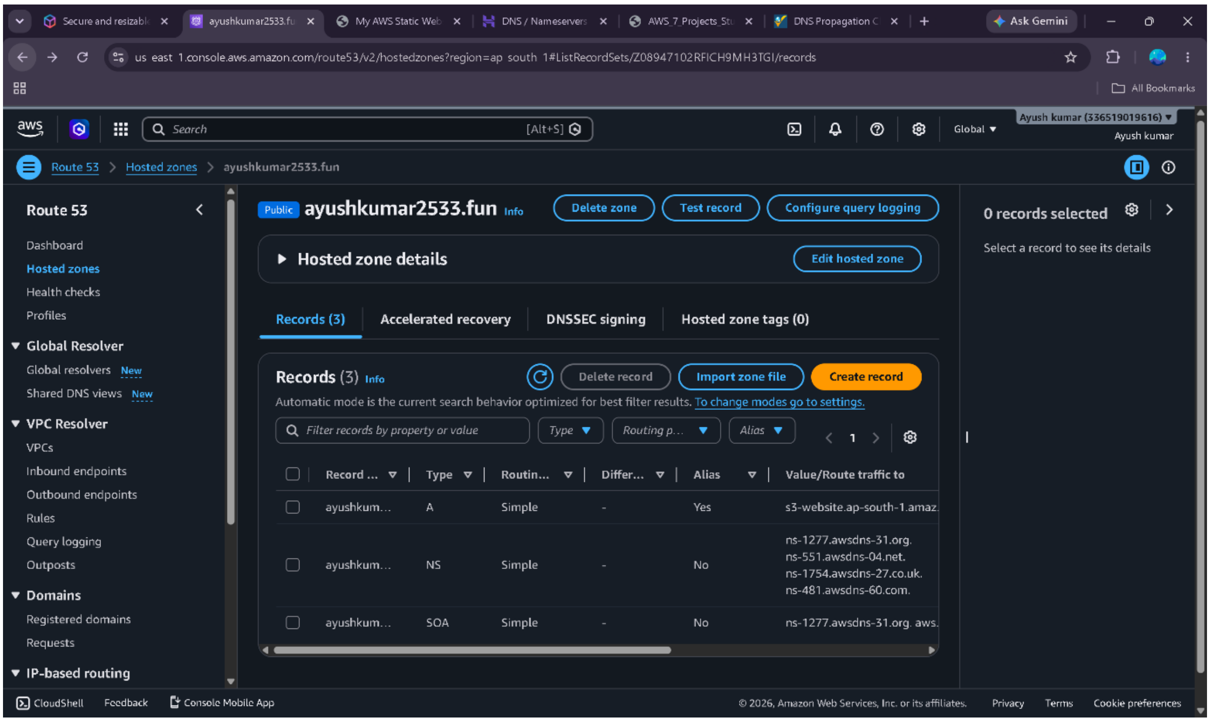
2. Part I: Domain Routing via AWS S3 Endpoint (Serverless)

2.1 Prerequisites

- A pre-configured AWS S3 static website.
- *Crucial Condition:* The S3 bucket name must **exactly match** the registered domain name (e.g., www.yourdomain.com) for the Route 53 Alias record to function properly.

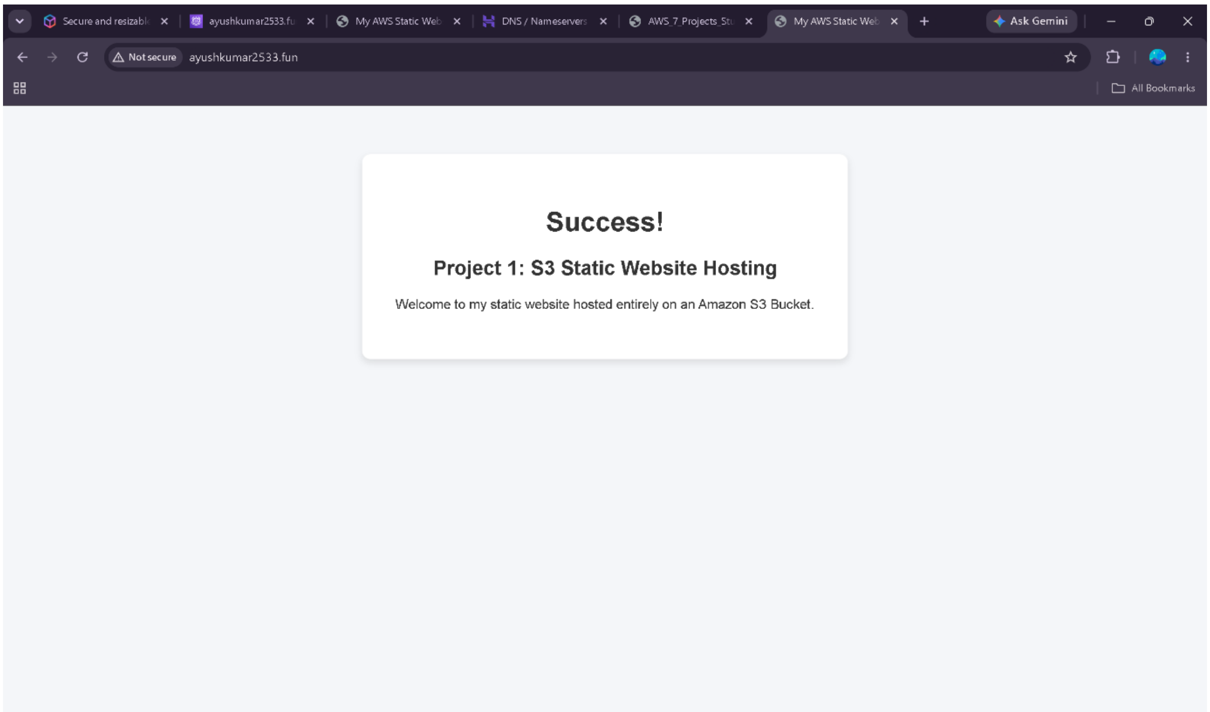
2.2 Step-by-Step Implementation

1. **Open Route 53 Dashboard:** Navigate to the AWS Route 53 Console and select **Hosted Zones**. Select your existing custom domain or register a new one.
2. **Create DNS Record:** Click on **Create Record**.
    - **Record Type:** Choose **A** record.
    - **Alias:** Toggle the switch to **Yes**.
    - **Route Traffic To:** Select *Alias to S3 website endpoint*.
    - **Region & Bucket:** Select the correct AWS Region (ap-south-1) and pick your matching S3 bucket.
3. **Propagation:** Save the configuration and allow 5–10 minutes for regional DNS propagation.



2.3 Verification & Live Test

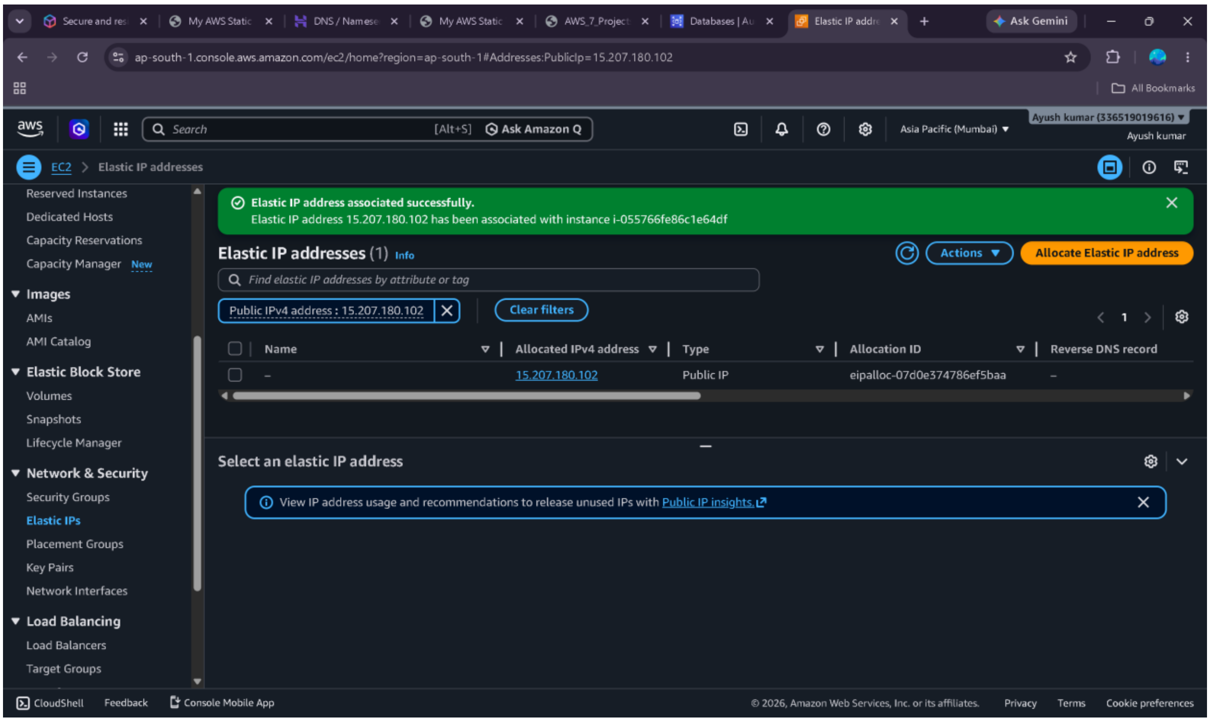
- Open a fresh web browser tab and type www.yourdomain.com.
- The browser will successfully render the static index.html file pulled straight from the S3 bucket without utilizing any traditional web server.

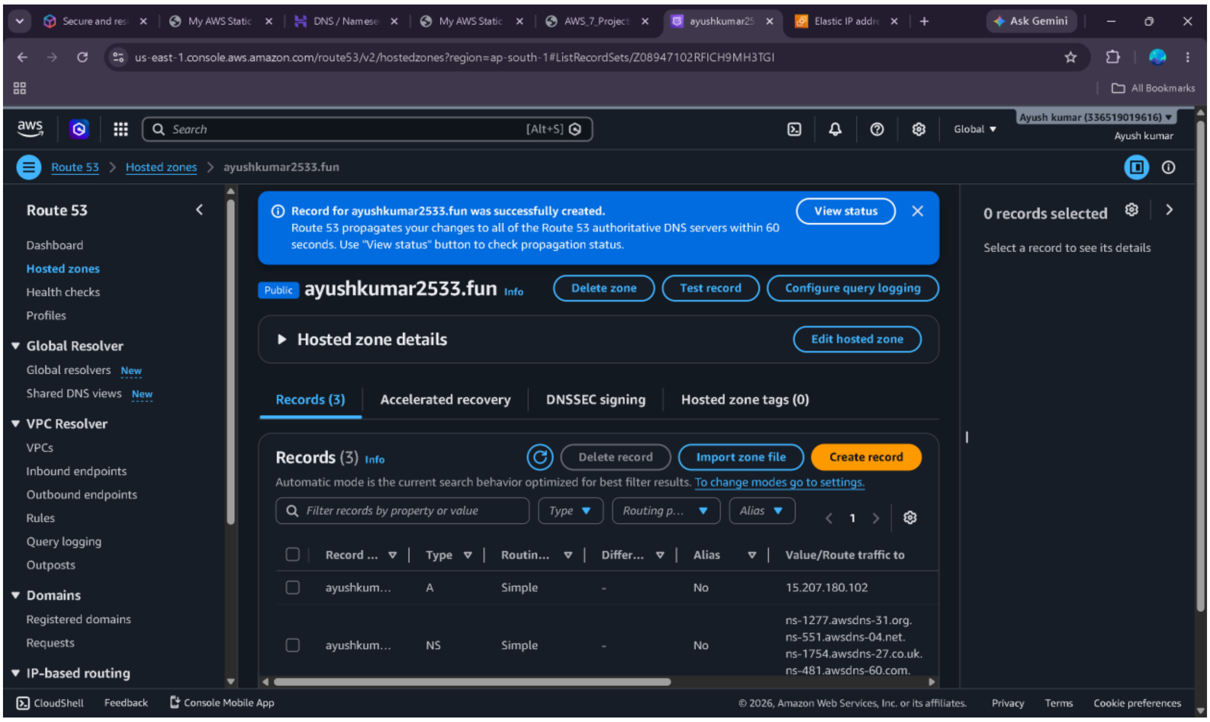


3. Part II: Domain Routing via AWS EC2 Public IP (Live Server)

3.1 Step-by-Step Implementation

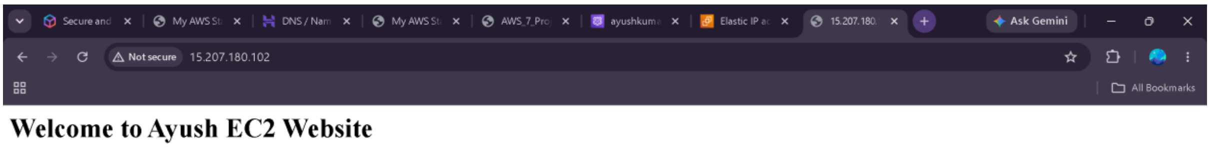
1. **Deploy EC2 Instance:** Provision an Amazon Elastic Compute Cloud (EC2) instance and install a web server package such as Apache (httpd) or Nginx.
2. **Allocate Elastic IP:** Allocate an **Elastic IP Address** from the Amazon pool and associate it directly with your active EC2 instance to ensure a static public IP.
3. **Update Route 53 Records:**
    - Navigate back to your Route 53 **Hosted Zone**.
    - Click **Create Record** and apply these specifications:
        - **Record Type:** A.
        - **Alias:** No.
        - **Value:** Input the assigned static **Elastic IP address** of your EC2 instance.
        - **TTL (Time To Live):** Set to 300 seconds.
4. Save the changes and wait briefly for the new records to update globally across DNS servers.





3.2 Verification & Live Test

- Launch a browser and navigate to yourdomain.com.
- The server traffic will hit the Route 53 DNS, resolve to the Elastic IP, and display the live active website served by Apache/Nginx.



4. Conclusion

This project successfully demonstrated the core capabilities of **AWS Route 53** as a highly reliable and scalable Domain Name System (DNS) service. By implementing both routing methods, a clear understanding of cloud infrastructure strategies was achieved:

- **Serverless Deployment (S3):** Routing traffic to an AWS S3 static website endpoint highlighted a cost-effective, high-availability architecture that completely eliminates server management overhead for static content.
- **Traditional Web Server Deployment (EC2):** Routing traffic to an AWS EC2 instance using a static Elastic IP showcased the workflow required for dynamic applications that demand dedicated compute resources.

Ultimately, the project provided hands-on experience in managing DNS records (A records and Alias records), understanding the nuances of DNS propagation, and executing essential cloud resource cleanup to enforce strict cost-optimization practices.